Chen, Zhaoting

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I study the large scale structure of the Universe, primarily using neutral hydrogen intensity mapping with MeerKAT and the future SKAO.

Employment

Henri Poincaré Early-Career Researcher Fellow 2026-
Observatoire de la Côte d'Azur, Université Côte d'Azur

Postdoctoral Research Associate 2023-2026
Institute for Astronomy, University of Edinburgh

Education

Ph.D. in Astrophysics 2020–2023
Jodrell Bank Centre for Astrophysics, University of Manchester Supervisor: Dr. Laura Wolz
Co-Supervisor: Prof. Richard Battye

- Thesis: "Interferometric Neutral Hydrogen Intensity Mapping in the Post-Reionization Universe"

M.Sc. in Astronomy and Astrophysics (Distinction) 2019–2020
Jodrell Bank Centre for Astrophysics, University of Manchester Supervisor: Dr. Laura Wolz
Co-Supervisor: Prof. Richard Battye

- Thesis: "Interferometric Neutral Hydrogen Intensity Mapping"

B.Sc. in Physics 2015–2019
University of Science and Technology of China (USTC) Supervisor: Prof. Yi-Fu Cai

Student Supervision

- **PhD co-supervision**
  - Boyan Zhao, PhD student from NAOC with Prof. Xuelei Chen [13] 2026-
  - Sefa Pamuk, visiting PhD student from IFCA with Dr. José Luis Bernal 2026-
  - Gabrielle Autieri, PhD student at SISSA with Dr. Marta Spinelli and Prof. Matteo Viel [1] 2025-
  - Jiakang Han, PhD student at UNITO (soon postdoc at KASI) with Prof. Stefano Camera 2024-2026
- **Project Supervisor**
  - Shailesh Devarasu, M.Sc. Theoretical Physics and Mathematical Physics at UoE 2026
  - Grace Tait, Astrophysics BSc at UoE 2026
  - Kyle Gillespie, Astrophysics BSc at UoE 2025

Publications

Search in: NASA ADS; INSPIRE-HEP

As the first and/or corresponding author:

1. G. Autieri, M. Spinelli and **Z. Chen** et al., "Towards constraining cosmology with HI–galaxy cross-correlations from MeerKLASS", submitted to *MNRAS*. arXiv: 2610.38733

2. **Z. Chen** and José Luis Bernal, "Emission line stacking on cosmological scales: equivalence to two-point statistics and modelling of the observational effects", submitted to *PRD*. arXiv: 2610.33603

3. **Z. Chen** and S. Cunnington et al. (MeerKLASS collaboration), "meer21cm: an Analysis Pipeline and Comprehensive Toolkit for HI Intensity Mapping", submitted to *OJA*. arXiv: 2607.01864

4. A. Mazumder and **Z. Chen** et al., "AASKA II: Interferometric HI Intensity Mapping of the Late Time Universe with SKA-Mid", *to appear*. arXiv: 2606.24730

5. **Z. Chen**, "A quadratic estimator view of the transfer function correction in intensity mapping surveys", *Mon.Not.Roy.Astron.Soc.Letters* 542 (2025) 1, L1-L6. arXiv: 2504.13070

6. **Z. Chen** et al. (MeerKLASS collaboration), "Emission line stacking of 21cm intensity maps with MeerKLASS: Inference pipeline and application to the L-band deep-field data", *ApJS* 279 (2025) 19. arXiv: 2504.03908

7. **Z. Chen**, and A. Pourtsidou, "Power spectrum multipoles and clustering wedges during the Epoch of Reionization", *Mon.Not.Roy.Astron.Soc.* 535 (2024) 3, 2228-2245. arXiv: 2405.05414

8. **Z. Chen**, E. Chapman, L. Wolz and A. Mazumder, "Detecting the HI Power Spectrum in the Post-Reionization Universe with SKA-Low", *Mon.Not.Roy.Astron.Soc.* 524 (2023) 3, 3724. arXiv: 2302.11504

9. **Z. Chen**, L. Wolz and R. Battye, "Towards Optimal Foreground Mitigation Strategies for Interferometric HI Intensity Mapping in the Low-Redshift Universe", *Mon.Not.Roy.Astron.Soc.* 518 (2023) 2, 2971–2990. arXiv: 2205.07776

10. **Z. Chen**, L. Wolz, M. Spinelli and S. G. Murray, "Extracting HI Astrophysics from Interferometric Intensity Mapping", *Mon.Not.Roy.Astron.Soc.* 502 (2021) 4, 5259–5276. arXiv: 2010.07985

11. **Z. Chen**, W. Luo, Y.-F. Cai, and E. Saridakis, "New test on general relativity and $f(T)$ torsional gravity from galaxy-galaxy weak lensing surveys", *Phys. Rev. D* 102 (2020), 104044. arXiv: 1907.12225

12. **Z. Chen**, Y. Xu, Y. Wang and X. Chen, "Stages of Reionization as revealed by the Minkowski Functionals", *Astrophys.J.* 885 (2019) 23. arXiv: 1812.10333

With significant contributions:

13. B. Zhao, E. Acedo and **Z. Chen** et al., "Mitigating oscillatory systematics in MeerKLASS L-band HI intensity maps", submitted to *MNRAS*. arXiv: 2611.13033

14. MeerKLASS collaboration (incl. **Z. Chen**), "HI intensity mapping with MeerKAT: A measurement of the 21-cm auto-spectrum on cosmological scales", *MNRAS* in press. arXiv: 2610.12158. [**Press release**]

15. L. Wolz et al. (incl. **Z. Chen**), "Advancing Astrophysics with the Square Kilometre Array (AASKA) II: Cosmology with HI Intensity Mapping", *to appear*. arXiv: 2607.03259

16. S. Majumdar et al. (incl. **Z. Chen**), "AASKA II: Cosmology with Intensity Mapping via Statistics Beyond the Power Spectrum in the SKAO Era", *to appear*. arXiv: 2606.30200

17. M. Santos, S. Camera, **Z. Chen**, S. Cunnington and J. Fonseca, "Cosmology with ESO–SKAO Synergies", *the Messenger*, 193 (2024) 20-23. arXiv: 2503.03915

18. A. Mazumder, L. Wolz, **Z. Chen** et al. "HI Intensity Mapping with the MIGHTEE Survey: First Results of the HI Power Spectrum", *Mon.Not.Roy.Astron.Soc.* 541 (2025) 1, 476-493. arXiv: 2501.17564

19. MeerKLASS collaboration (incl. **Z. Chen**), "MeerKLASS L-band deep-field intensity maps: entering the HI dominated regime", *Mon.Not.Roy.Astron.Soc.* 537 (2025) 4, 3632-3661. arXiv: 2407.21626

20. K. Diao, **Z. Chen**, X. Chen and Y. Mao, "Reionization Parameter Inference from 3D Minkowski Functionals of the 21 cm Signals", *Astrophys.J.* 974 (2024) 141. arXiv: 2406.20058

21. S. Paul, **Z. Chen**, M. G. Santos, and L. Wolz, "A Direct Detection of Neutral Hydrogen Intensity Mapping on Mpc Scales at $z \approx 0.32$ and $z \approx 0.44$", *Astrophys.J.Letters* 1005 (2026) 2, L56. arXiv: 2301.11943. [**Press release**]

22. S. G. Murray, B. Diemer, **Z. Chen** et al., "TheHaloMod: An online calculator for the halo model", *Astron.Comput.* 36 (2021) 100487. arXiv: 2009.14066

23. B. Li, **Z. Chen**, Y.-F. Cai, and Y. Mao, "Testing the scale-dependent hemispherical asymmetry with the 21-cm power spectrum from the epoch of reionization", *Mon.Not.Roy.Astron.Soc.* 487 (2019) 4, 5564-5571. arXiv: 1904.04683

As part of a collaboration:

24. D. Tassie et al. (incl. Z. Chen), "Detecting redshift space distortions with forthcoming 21cm intensity maps from MeerKLASS", submitted to *MNRAS*. arXiv: 2610.13393

25. J. Chaves-Montero et al. (incl. Z. Chen), "Lyman-α forest holography: 3D predictions from 1D measurements", submitted to *A&A*. arXiv: 2607.27413

26. M. F. Ruiz-Herrera Bernal et al. (incl. Z. Chen), "CoLoRe-2LPT: Lyman-α mock catalogues for the validation of DESI cosmological analyses", submitted to *A&A*. arXiv: 2607.27412

27. M. Herbold et al. (incl. Z. Chen), "Validation of the DESI DR2 Lyα forest full-shape analysis", submitted to *PRD*. arXiv: 2607.27411

28. DESI collaboration (incl. Z. Chen), "DESI DR2 Results IV: Alcock-Paczyński Measurements from the Lyman Alpha Forest and Cosmological Constraints", submitted to *PRD*. arXiv: 2607.27410

29. M. Spinelli et al. (incl. Z. Chen), "AASKA II: Methodological Frontiers in 21-cm Intensity Mapping: the Treatment of Systematics and Foreground Contamination", *to appear*. arXiv: 2606.27244

30. S. Cunnington and J. Wang (incl. Z. Chen), "AASKA II: Single-dish HI Intensity Mapping with the SKAO: Precursor Progress with MeerKAT's Large Area Synoptic Survey (MeerKLASS)", *to appear*. arXiv: 2606.25955

31. G. Chen et al. (incl. Z. Chen), "Transformer-based Source Detection and Morphological Classification in LOFAR Deep Fields Continuum Images", *Mon.Not.Roy.Astron.Soc.* 549 (2026), 3. arXiv: 2605.30500

32. J. Townsend et al. (incl. Z. Chen), "Measurements of the HI intensity mapping power spectrum at low redshifts with MIGHTEE data: comparison with detected HI galaxies", submitted to *MNRAS*. arXiv: 2602.23055

33. S. Cunnington et al. (incl. Z. Chen), "Revealing cosmological fluctuations in 21cm intensity maps with MeerKLASS: from maps to power spectra", the Springer Nature Astronomy Prize Awardees Collections (2025), *Astrophys. Space Sci.* 371 (2026) 2, 16. arXiv: 2510.27549

34. A. Bonaldi et al. (incl. Z. Chen), "Square Kilometre Array Science Data Challenge 3a: foreground removal for an EoR experiment", *Mon.Not.Roy.Astron.Soc.* 543 (2025) 2, 1092-1119. arXiv: 2503.11740

Academic Service

- **Referee for MNRAS, ApJ, PRL, OJA, A&A** 2022-
- **LOC member for Higgs Centre workshop** 2026

Collaboration Roles

MeerKLASS collaboration
- Co-chair of the power spectrum working group

SKAO Cosmology Science Working Group
- Co-lead of the HI intensity mapping focus group

Dark Energy Spectroscopic Instrument
- Member of the C3 working group working on cross-correlation with 21 cm line.
- Member of the GQC working group working on full shape analysis.

Media

- **S. Paul, Z. Chen, M. Santos and L. Wolz (2026) appeared in:**
  - Press releases from University of Manchester [link], University of Western Cape [link], and South African Radio Astronomy Observatory [link]
  - Selected news articles from Xinhua [link], Universe Today [link], and Space.com [link]
- **MeerKLASS collaboration (incl. Z. Chen) (2026) appeared in:**
  - Press releases from South African Radio Astronomy Observatory [link],

Selected Talks

21cm & Cosmological Lines 2026 (invited) 10/2026
Title: Hands-on look into MeerKLASS with MEER21CM INPE, Brazil (online)

Mining the rainbow: Cosmology in Multicolour via LIM surveys (invited) 07/2025
Title: Towards robust inference of intensity mapping power spectrum SISSA

A New Era in Astrophysics: preparing for early science with the SKAO 06/2025
Title: Emission line stacking of neutral hydrogen intensity maps: results from MeerKAT Görlitz, Germany

CASTLE 2024 09/2024
Title: Neutral Hydrogen stacking from MeerKLASS Tagliolo Monferrato, Italy

Tianlai 2024 07/2024
Title: Clustering wedges of power spectrum multipoles during Epoch of Reionization HDU, Hangzhou

Moriond 2024 04/2024
Title: Clustering wedges of power spectrum multipoles during Epoch of Reionization La Thuile, Italy

UK National Astronomy Meeting 2023 07/2023
Title: A first detection of neutral hydrogen intensity mapping Cardiff University

National Astronomical Observatories of China seminar 04/2023
Title: A first detection of neutral hydrogen intensity mapping on Mpc scales at $z \approx 0.32$ and $z \approx 0.44$ NAOC

SKAO Cosmology Science Working Group meeting 2023 01/2023
Title: A first detection of neutral hydrogen intensity mapping on Mpc scales at $z \approx 0.32$ and $z \approx 0.44$ JBCA

HITS (HI Intensity Mapping in Trieste) 2022 05/2022
Title: Interferometric Intensity Mapping SISSA Trieste

SAZERAC 21cm 2022 03/2022
Title: Interferometric Intensity Mapping in the Low-Redshift Universe Online

UK National Astronomy Meeting 2021 07/2021
Title: Extracting HI Astrophysics from Interferometric Intensity Mapping University of Bath

2021 SKA Science Conference 03/2021
Title: Extracting HI Astrophysics from Interferometric Intensity Mapping SKA Organisation

SWIFAR Colloquium (invited) 09/2020
Title: Halo Model, Interferometric Intensity Mapping and HI Shot Noise Yunnan University